

2. Duchenne muscular dystrophy (DMD) is a recessive sex-linked condition which affects 1 in 3 500 boys. It is caused by a mutation of a gene which codes for the protein dystrophin, which gives structural stability to all membranes within muscle cells. As a result, calcium ions enter the mitochondria and they burst. This results in the death of the muscle cell. The mutation involves a deletion involving an intron and one or more exons.

(a) (i) Explain why a mutation involving an intron would not affect the primary structure of a polypeptide but a mutation in an exon might. [2]

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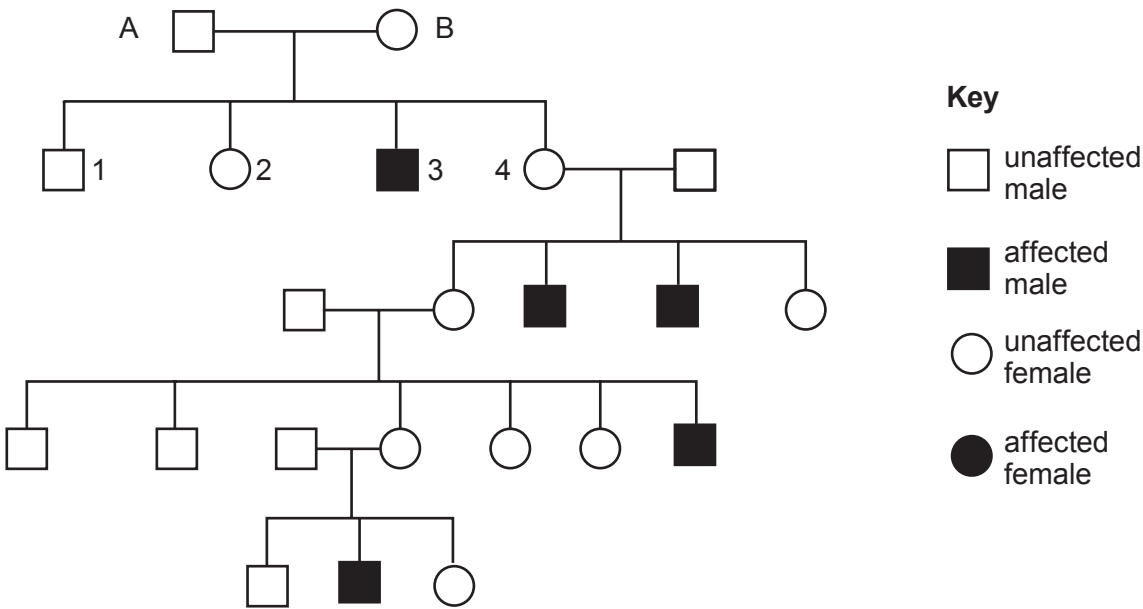
(ii) Explain why the mitochondria burst after calcium ions enter them. [2]

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(b) The pedigree chart below shows the transmission of DMD in humans from one generation to another.



- (i) Using the symbols X^D for unaffected and X^d for affected by DMD to represent alleles, construct a genetic diagram to show the genotypes and phenotypes of the parents A and B and the possible genotypes of their offspring. [4]

Parental phenotypes
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Parental genotypes
Gametes

Offspring genotypes

- (ii) Explain why it is not possible to determine the genotype of **child 2**. [1]

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- (c) Through gene therapy, it is hoped that the functional version of the gene can be isolated and introduced into muscle cells. Historically, a virus has been used as the vector.

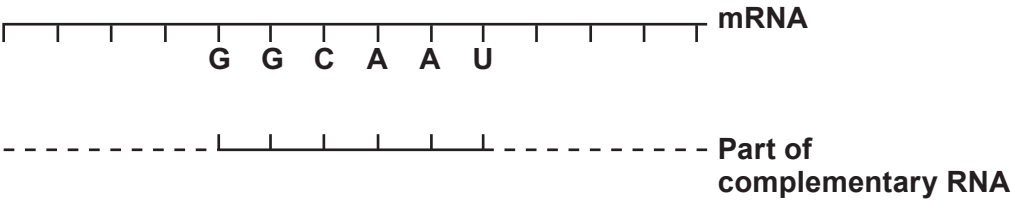
Suggest **two** potential problems of using a virus to treat DMD. [2]

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(d) DMD can be treated using a drug which acts as a ‘molecular patch’ on an mRNA molecule transcribed from the mutated DNA. The drug contains a short RNA molecule which is complementary to a specific sequence of bases on the mRNA. It prevents translation of the covered mRNA but allows ribosomes to continue translating the strand following the patch.

(i) **Complete the diagram** below to show the complementary base pairing between the mRNA and the molecular patch. [1]



(ii) State **one** difference between the structure of the protein dystrophin synthesised after the use of the molecular patch and normal dystrophin. [1]

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(iii) There is another drug available which is used to remove mutated exons from the dystrophin gene. State the advantage of using this drug to treat DMD rather than the molecular patch on mRNA. [1]

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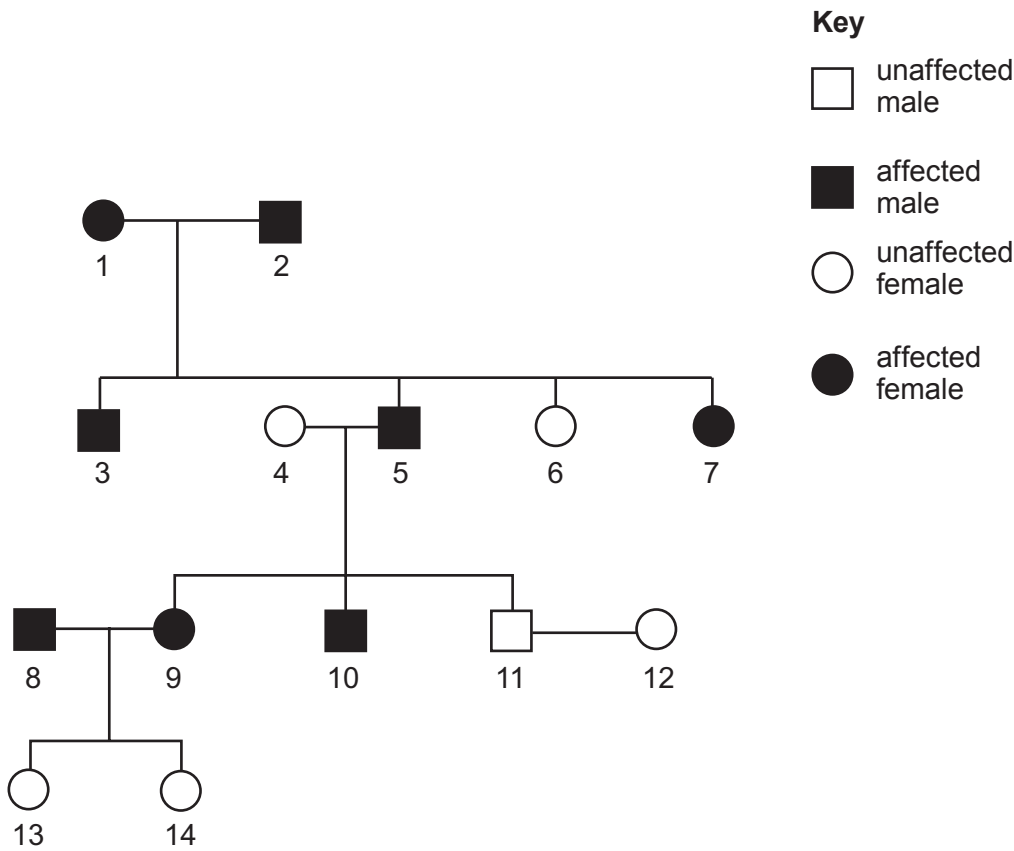
(iv) Scientists are also investigating a means of using a molecular patch for germ line gene therapy. State **one** ethical issue of using germ line gene therapy in the treatment of DMD. [1]

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(e) The pedigree chart below shows the transmission of another form of muscular dystrophy caused by a gene mutation.



What conclusions can be made about the inheritance of this form of muscular dystrophy? **Giving 3 specific examples** from the pedigree chart, explain how you arrived at your conclusions. [5]

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